

# Supplementary Material for CAPSNet: Calibrating Proposal-Score Reliability for Weakly Supervised Temporal Action Localization

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## 1 Notation Summary

Table 1 collects the notation used throughout the Method section. Indices  $i, j$  refer to proposals,  $k$  to ordered RoI bins, and  $c$  to action classes.

**Table 1.** Core notation.

| Symbol                                          | Meaning                                                                                  |
|-------------------------------------------------|------------------------------------------------------------------------------------------|
| $p_i = (t_i^s, t_i^e)$                          | Temporal proposal used as the detection unit.                                            |
| $F_i \in \mathbb{R}^{K \times D}$               | Ordered RoI-bin representation of proposal $p_i$ .                                       |
| $z_i^b, a_i, u_i$                               | Base classification logits, proposal-attention logit, and completeness logit.            |
| $h_{i,k,c}$                                     | Bin-level class logit for class $c$ .                                                    |
| $o_{i,k,c}$                                     | Class-aware action-core proportion for bin $k$ and class $c$ .                           |
| $o_{i,k}^{loc}$                                 | Class-agnostic localization proportion for bin $k$ .                                     |
| $z_{i,c}^{core}, z_{i,c}^{ctx}$                 | Action-core and contextual evidence for the same action class.                           |
| $d_{i,c}$                                       | Action dominance, measuring whether action-core evidence exceeds context.                |
| $\chi_{i,c}^{cap}, b_{i,c}, \omega_{i,c}^{res}$ | CAP-side action-core quality, Base confidence, and rescue gate.                          |
| $z_{i,c}^{cap}, z_{i,c}^{ref}$                  | CAP branch logits and Base-CAP combined logits.                                          |
| $\hat{a}_i, \hat{u}_i$                          | Sigmoid proposal attention and CAP-refined completeness.                                 |
| $w_{ijc}, w_{ic}, w_{i,c}$                      | Detached weights for CAP rescue pairs, CAP dominance pairs, and CPS anchors.             |
| $q_i$                                           | CPS query representation for proposal $p_i$ .                                            |
| $v_{i,c}$                                       | Indicator that class $c$ has at least one valid cross-video support for proposal $p_i$ . |
| $\alpha_{i,j}^c, r_{i,c}$                       | Retrieval weight and temperature-weighted support prototype.                             |
| $\psi_{i,c}, \beta_{i,c}$                       | Retrieved support alignment and support confidence.                                      |
| $\rho_{i,c}$                                    | CPS statistic vector passed to the logit editor.                                         |
| $z_{i,c}^{cps}$                                 | CPS-edited action logit.                                                                 |
| $S_{i,c}$                                       | Final proposal score before Soft-NMS.                                                    |

## 2 Training Objective Details

### 2.1 Common Top- $k$ MIL Form

We use  $\text{sg}(\cdot)$  for stop-gradient,  $[x]_+ = \max(x, 0)$ , and  $\epsilon$  for a small numerical constant. For proposal logits  $z \in \mathbb{R}^{M \times (C+1)}$ , where  $M$  is the number of proposals

and  $C$  is the number of action classes, the video-level logit of class  $c$  is

$$\bar{z}_c(z) = \frac{1}{K_t} \sum_{i \in \text{TopK}_{K_t}(z, c)} z_{i,c}. \quad (1)$$

$K_t$  denotes the top- $k$  pooling size, which is set to 8 in Table 3. The normalized weak label is  $\tilde{y}^{(\xi)} = [y, \xi] / (\sum_{c=1}^C y_c + \xi + \epsilon)$ , where  $y$  is the video-level multi-hot label. Setting  $\xi = 1$  assigns a weak background target, while  $\xi = 0$  leaves background only as a competing softmax class. The generic MIL loss is

$$\mathcal{L}_{\text{mil}}(z, y; \xi) = - \sum_{c=1}^{C+1} \tilde{y}_c^{(\xi)} \log \text{softmax}(\bar{z}(z))_c. \quad (2)$$

## 2.2 Base Proposal MIL Objective

The main paper abbreviates the baseline objective as  $\mathcal{L}_{\text{base}}$  so the Method section can focus on the reliability modules. Training still optimizes this term together with CAP and CPS. Following the proposal-level MIL baseline in Sec. 3.1, each RGB or flow branch  $m \in \{r, f\}$  produces proposal logits  $z^{b,m}$ , proposal attention  $a^m$ , completeness logits  $u^m$ , and surrounding-contrast logits  $z^{s,m}$ . The surrounding branch belongs to the Base scorer rather than CAPSNet. The objective is

$$\begin{aligned} \mathcal{L}_{\text{base}} = & \frac{1}{2} \sum_{m \in \{r, f\}} [w_{\text{orig}} \mathcal{L}_{\text{mil}}(z^{b,m}, y; 1) + w_{\text{supp}} \mathcal{L}_{\text{mil}}(z^{s,m}, y; 0)] \\ & + w_{\text{irc}} \mathcal{L}_{\text{irc}} + w_{\text{comp}} \mathcal{L}_{\text{comp}}. \end{aligned} \quad (3)$$

We use  $w_{\text{orig}} = 2$ ,  $w_{\text{supp}} = 1$ ,  $w_{\text{irc}} = 2$ , and  $w_{\text{comp}} = 20$ .

High-attention pseudo instances define the completeness target. For branch  $m$ , NMS over proposals satisfying  $a_i^m > \gamma \max_j a_j^m$  gives the pseudo-instance set  $\mathcal{S}^m$ , with attention threshold  $\gamma = 0.8$ . For the valid proposal set  $\Omega$ , the target coverage is

$$\hat{\gamma}_i^m = \max_{p_j \in \mathcal{S}^m} \text{tIoU}(p_i, p_j). \quad (4)$$

The completeness loss is

$$\mathcal{L}_{\text{comp}} = \frac{1}{2} \sum_{m \in \{r, f\}} \frac{1}{|\Omega|} \sum_{i \in \Omega} (\sigma(u_i^m) - \hat{\gamma}_i^m)^2. \quad (5)$$

$\mathcal{L}_{\text{irc}}$  enforces cross-stream proposal-rank consistency. For each present class, one stream matches the other stream’s proposal ranking with KL divergence over temporally overlapping proposals; low-attention proposals are ignored.

## 2.3 CAP Auxiliary Terms

CAP uses the action-core/context decomposition from the main paper. Let  $\bar{o}_i^{\text{loc}} = \frac{1}{K} \sum_{k=1}^K o_{i,k}^{\text{loc}}$  be the proposal-level localization proportion. The action-core quality and Base confidence are

$$\chi_{i,c}^{\text{cap}} = [\sigma(z_{i,c}^{\text{core}}) - \lambda_{\text{ctx}} \sigma(z_{i,c}^{\text{ctx}}) + \lambda_{\text{occ}} \bar{o}_i^{\text{loc}}]_+. \quad (6)$$

$$b_{i,c} = \sigma(a_i) \sigma(u_i) \text{softmax}(z_i^b)_c. \quad (7)$$

$$\omega_{i,c}^{res} = [\chi_{i,c}^{cap} - b_{i,c}]_+ ((1 - \lambda_{amb}) + \lambda_{amb} 4b_{i,c}(1 - b_{i,c})). \quad (8)$$

The coefficients  $\lambda_{ctx}$ ,  $\lambda_{occ}$ , and  $\lambda_{amb}$  are listed in Table 3. The final correction gate is  $\omega_{i,c} = g_{i,c} + \omega_{i,c}^{res}$ , where  $g_{i,c}$  is the CAP confidence-prior gate. We clip the sum to keep the residual bounded. In the losses below,  $\hat{a}_i = \sigma(a_i)$  and  $\hat{u}_i$  denotes the CAP-refined completeness score.

CAP localization reuses the Base pseudo-instance construction and trains the CAP-refined completeness estimate. The target  $\hat{\gamma}_i$  is the branch-averaged coverage:

$$\mathcal{L}_{\text{cap-loc}} = \frac{1}{|\Omega|} \sum_{i \in \Omega} (\hat{u}_i - \hat{\gamma}_i)^2. \quad (9)$$

CAP discrimination uses a positive-recovery ranking term and an action-dominance term. We define

$$\pi_{i,c}^{cap} = \text{softmax}(z_i^{cap})_c, \quad \pi_{i,c}^{ref} = \text{softmax}(z_i^{ref})_c. \quad (10)$$

The rescue set  $\mathcal{R}$  contains triplets  $(i, j, c)$  in which proposal  $i$  has positive disagreement,  $\chi_{i,c}^{cap} > b_{i,c}$ , and proposal  $j$  is a harder non-rescue candidate,  $b_{j,c} > \chi_{j,c}^{cap}$ . The detached pair weight is

$$\begin{aligned} w_{ijc} &= \text{sg} \left( \sqrt{\nu_{i,c}^+ \nu_{j,c}^-} \right), \\ \nu_{i,c}^+ &= [\chi_{i,c}^{cap} - b_{i,c}]_+ \chi_{i,c}^{cap}, \\ \nu_{j,c}^- &= [b_{j,c} - \chi_{j,c}^{cap}]_+ \frac{b_{j,c} + \hat{a}_j \hat{u}_j}{2}. \end{aligned} \quad (11)$$

This weight emphasizes pairs with a reliable rescue proposal and a hard non-rescue proposal. The rescue loss with margin  $m_r$  is

$$\mathcal{L}_{\text{cap-res}} = \frac{1}{Z_r} \sum_{(i,j,c) \in \mathcal{R}} w_{ijc} [m_r - \pi_{i,c}^{cap} + \pi_{j,c}^{cap}]_+. \quad (12)$$

For action dominance,  $\Omega^+$  contains the top-scoring pseudo-positive proposal-class pairs selected by  $\hat{a}_i \hat{u}_i \pi_{i,c}^{ref}$  over present classes. With  $w_{ic} = \text{sg}(\hat{a}_i \hat{u}_i \pi_{i,c}^{ref})$ , the dominance loss with margin  $m_d$  is

$$\mathcal{L}_{\text{cap-dom}} = \frac{1}{Z_d} \sum_{(i,c) \in \Omega^+} w_{ic} [m_d - d_{i,c}]_+. \quad (13)$$

The discrimination loss sums the rescue and dominance terms:

$$\mathcal{L}_{\text{cap-disc}} = \mathcal{L}_{\text{cap-res}} + \mathcal{L}_{\text{cap-dom}}. \quad (14)$$

Its normalizers are  $Z_r = \sum_{(i,j,c) \in \mathcal{R}} w_{ijc} + \epsilon$  and  $Z_d = \sum_{(i,c) \in \Omega^+} w_{ic} + \epsilon$ .

## 2.4 CPS Loss Terms

CPS-MIL applies the common top- $k$  MIL loss to the support-edited logits:

$$\mathcal{L}_{\text{cps-mil}} = \mathcal{L}_{\text{mil}}(z^{\text{cps}}, y; 0). \quad (15)$$

In the support losses,  $\beta_{i,c}$  is the support confidence predicted by the CPS editor, and  $\psi_{i,c}$  is the retrieved support-alignment score defined in the support-bank subsection. Let  $\mathcal{C}^+(v)$  and  $\mathcal{C}^-(v)$  be the present and absent classes of video  $v$ . Thus  $y_{v,c} = 1$  for  $c \in \mathcal{C}^+(v)$ , and  $y_{v,c} = 0$  for  $c \in \mathcal{C}^-(v)$ . Pseudo-positive anchors for present class  $c$  are selected by CAP-refined proposal confidence:

$$s_{i,c}^{\text{ref}} = \text{softmax}(z_i^{\text{ref}})_c \sigma(a_i) \hat{u}_i. \quad (16)$$

For the valid proposal set  $\Omega$ , the anchor count is  $K_a = \max(1, \lfloor |\Omega|/8 \rfloor)$ . The anchor set and detached anchor weight are

$$A_{v,c}^+ = \text{TopK}_{K_a}(s_{:,c}^{\text{ref}}). \quad (17)$$

$$w_{i,c} = \text{sg}(s_{i,c}^{\text{ref}}). \quad (18)$$

For each anchor,  $\mathcal{C}_i^-(v) = \{c' \in \mathcal{C}^-(v) \mid v_{i,c'} = 1\}$  contains absent classes with valid support. If this set is empty, the loss skips the absent-support term. Define

$$\ell_i^- = \begin{cases} \frac{1}{|\mathcal{C}_i^-(v)|} \sum_{c' \in \mathcal{C}_i^-(v)} \log(1 - \beta_{i,c'}), & |\mathcal{C}_i^-(v)| > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

The support supervision loss is

$$\mathcal{L}_{\text{cps-sup}} = -\frac{1}{Z_v} \sum_{c \in \mathcal{C}^+(v)} \sum_{i \in A_{v,c}^+} w_{i,c} (v_{i,c} \log \beta_{i,c} + \ell_i^-), \quad (20)$$

where  $Z_v = \sum_{c \in \mathcal{C}^+(v)} |A_{v,c}^+| + \epsilon$ . Let  $A_{v,c}^{\text{comp}} = \{i \in A_{v,c}^+ \mid |\mathcal{C}_i^-(v)| > 0\}$ . The present-vs-absent competition term uses margin  $m_s = 0.08$ :

$$\begin{aligned} \mathcal{L}_{\text{cps-comp}} = \frac{1}{Z_v} \sum_{c \in \mathcal{C}^+(v)} \sum_{i \in A_{v,c}^{\text{comp}}} w_{i,c} v_{i,c} \frac{1}{2} & \left( [m_s - \beta_{i,c} + \max_{c' \in \mathcal{C}_i^-(v)} \beta_{i,c'}]_+ \right. \\ & \left. + [m_s - \psi_{i,c} + \max_{c' \in \mathcal{C}_i^-(v)} \psi_{i,c'}]_+ \right). \end{aligned} \quad (21)$$

## 3 Implementation Details

### 3.1 Data Flow and Detach Rules

CAPSNet keeps a single proposal interface. External S-MIL detections provide temporal boundaries only. Base, CAP, and CPS then produce  $z^b, a, u, z^{\text{ref}}, \hat{u}$ , and  $z^{\text{cps}}$  in sequence.

All losses act on proposal logits rather than proposal-source S-MIL scores. We detach pseudo labels, pseudo-positive anchor weights, rescue-pair weights, stored support confidences, and stored support features. CPS updates the current query encoder and editor, but older support-bank entries remain fixed. During warm-up, detached Base/CAP predictions can populate the bank before CPS residual MIL is enabled. At inference, the model queries the frozen training-video bank; video-level scores only choose active classes.

### 3.2 External Proposal Boundary Interface

CAPSNet treats external S-MIL detections as boundary candidates. For each candidate segment, the model loads only the temporal start and end coordinates; proposal-source class labels and confidence scores are discarded before classification and calibration. Table 3 reports the proposal extension ratio, number of RoI bins, and training-time proposal cap.

### 3.3 Ordered-Bin Mixer in CAP

Before predicting action-core proportions, CAP applies a lightweight ordered-bin mixer that enriches RoI-bin features while preserving proposal coordinates and bin order. Table 2 gives the configuration. CAP heads then use the mixed tokens to predict bin-level class logits, class-aware action-core proportions, localization proportions, confidence-prior gates, and bounded residuals.

**Table 2.** Ordered-bin mixer configuration.

| Component   | Configuration                                                                          |
|-------------|----------------------------------------------------------------------------------------|
| Input       | $K \times D_e$ RoI-bin tokens, with $K = 12$ .                                         |
| Local path  | LayerNorm, depthwise Conv1D with kernels 3 and 5, concatenation, and pointwise Conv1D. |
| Token path  | LayerNorm and MLP over the ordered-bin dimension, with hidden size $\max(2K, 16)$ .    |
| Branch gate | MLP from pooled local/token summaries to two softmax weights.                          |
| Output      | Weighted local/token sum, residual projection, and addition to the input tokens.       |

### 3.4 Contextual Proposal Query

For proposal  $p_i$ , CPS builds the query  $q_i$  from the normalized proposal feature, the normalized temporal span, proposal confidence, and neighboring-proposal context. The proposal confidence is  $s_i = \hat{a}_i \hat{u}_i$ , and the neighboring context summarizes nearby proposal scores and temporal layouts. A lightweight MLP maps these inputs to a 256-dimensional query embedding. The query is used for support retrieval and logit editing, while the proposal boundary remains unchanged.

### 3.5 CPS Support Bank

CPS maintains one positive support bank  $M_c$  per action class. Each entry stores a normalized proposal feature, normalized temporal span, detached confidence, and source video id. Training videos update the bank, which is frozen at evaluation. Evaluation proposals are only queries, and same-video entries are masked during retrieval.

For a query proposal and class  $c$ , CPS retrieves up to  $K_m = 4$  valid supports and builds a temperature-weighted prototype with  $\tau = 0.07$ . If no valid support exists,  $v_{i,c} = 0$  and CPS leaves that class logit unchanged.

For support proposal  $m_j^c \in M_c$ , let  $e_j^c$ ,  $\ell_j^c$ ,  $\eta_j^c$ , and  $\nu_j^c$  denote its embedding, span, confidence, and source-video id. The query span and video id are  $\ell_i$  and  $\nu_i$ . Same-video entries are removed by  $\mathcal{I}_{i,c} = \{m_j^c \in M_c \mid \nu_j^c \neq \nu_i\}$ . CPS selects the support set  $\mathcal{N}_{i,c}$  by applying top- $K_m$  retrieval to  $\{s(q_i, e_j^c) : m_j^c \in \mathcal{I}_{i,c}\}$ . For  $m_j^c \in \mathcal{N}_{i,c}$ , the retrieval weight is

$$\alpha_{i,j}^c = \frac{\exp(s(q_i, e_j^c)/\tau)}{\sum_{m_\ell^c \in \mathcal{N}_{i,c}} \exp(s(q_i, e_\ell^c)/\tau)}. \quad (22)$$

Here  $s(\cdot, \cdot)$  is cosine similarity. The support prototype and averaged metadata are

$$\begin{aligned} r_{i,c} &= \sum_{m_j^c \in \mathcal{N}_{i,c}} \alpha_{i,j}^c e_j^c, \\ \bar{s}_{i,c} &= \frac{1}{|\mathcal{N}_{i,c}|} \sum_{m_j^c \in \mathcal{N}_{i,c}} s(q_i, e_j^c), \\ \bar{\eta}_{i,c} &= \sum_{m_j^c \in \mathcal{N}_{i,c}} \alpha_{i,j}^c \eta_j^c, \quad \bar{\ell}_{i,c} = \sum_{m_j^c \in \mathcal{N}_{i,c}} \alpha_{i,j}^c \ell_j^c. \end{aligned} \quad (23)$$

These quantities form the statistic vector passed to the CPS editor:

$$\rho_{i,c} = [\psi_{i,c}, \bar{s}_{i,c}, \bar{\eta}_{i,c}, -|\ell_i - \bar{\ell}_{i,c}|, s_i, \mu_i, \kappa_i], \quad (24)$$

Here  $\psi_{i,c} = s(q_i, r_{i,c})$  is the support-alignment score, and  $s_i = \hat{a}_i \hat{u}_i$  is the scalar proposal confidence. The term  $\mu_i = \max_{c'} \text{softmax}(z_i^{ref})_{c'}$  is the maximum action probability, and  $\kappa_i$  is the confidence aggregated from neighboring proposals by the contextual query encoder.

The support bank uses 64 FIFO slots per class and a 256-dimensional query embedding. EMA updates the target encoder for support construction with momentum 0.995; stored support confidences are detached CAP-refined scores.

## 4 Reproducibility Details

Table 3 lists the implementation and inference settings. THUMOS14 and ActivityNet v1.3 use the same module definitions unless noted otherwise.

**Table 3.** Main reproducibility settings.

| Setting                      | Value                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Input feature                | Two-stream I3D, 2048 dimensions.                                                                                           |
| RoI bins                     | $K = 12$ .                                                                                                                 |
| Maximum proposals per video  | 1000 during training.                                                                                                      |
| Proposal extension           | 25% duration on both temporal sides.                                                                                       |
| Optimizer                    | Adam.                                                                                                                      |
| Learning rate / weight decay | $5 \times 10^{-5}$ / $10^{-3}$ .                                                                                           |
| Dropout                      | 0.5.                                                                                                                       |
| Top- $k$ MIL pooling         | $K_t = 8$ ; pseudo-instance threshold $\gamma = 0.8$ .                                                                     |
| CAP coefficients             | $\lambda_{ctx} = 0.5$ , $\lambda_{occ} = 0.2$ , $\lambda_{amb} = 0.7$ , $\lambda_{ref} = 0.2$ ,<br>$\lambda_{bin} = 0.3$ . |
| CAP margins                  | $m_r = 0.05$ , $m_d = 0.1$ .                                                                                               |
| CPS retrieval                | $K_m = 4$ , $\tau = 0.07$ , $m_s = 0.08$ , 256-D queries, 64 slots per class.                                              |
| CPS support bank             | EMA momentum 0.995, enabled after a 20-epoch CPS-stage warm-up.                                                            |
| CPS pseudo-positive anchors  | Top $\max(1, \lfloor  \Omega /8 \rfloor)$ valid proposals per present class.                                               |
| Proposal score               | No video-level score is added to the final proposal score.                                                                 |
| Post-processing              | Soft-NMS.                                                                                                                  |

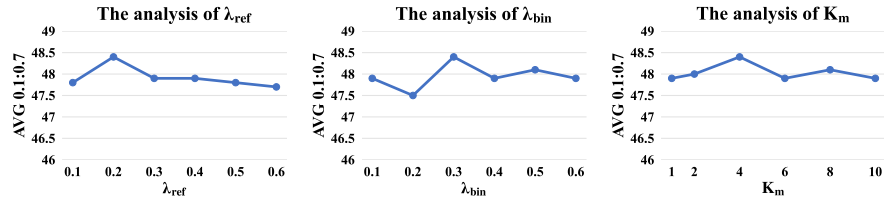
## 5 Hyperparameter Sensitivity

Table 4 reports the sensitivity analysis from the main paper. Each row changes one setting while keeping the others at their defaults.

**Table 4.** Sensitivity analysis of the main CAP/CPS hyperparameters on THUMOS14. AVG is over IoU thresholds 0.1:0.1:0.7.

| Exp | $\lambda_{ref}$ | $\lambda_{bin}$ | $K_m$ | AVG<br>0.1:0.7 |
|-----|-----------------|-----------------|-------|----------------|
| 1   | 0.1             | 0.3             | 4     | 47.8           |
| 2   | 0.3             | 0.3             | 4     | 47.9           |
| 3   | 0.4             | 0.3             | 4     | 47.9           |
| 4   | 0.5             | 0.3             | 4     | 47.8           |
| 5   | 0.6             | 0.3             | 4     | 47.7           |
| 6   | 0.2             | 0.1             | 4     | 47.9           |
| 7   | 0.2             | 0.2             | 4     | 47.5           |
| 8   | 0.2             | 0.4             | 4     | 47.9           |
| 9   | 0.2             | 0.5             | 4     | 48.1           |
| 10  | 0.2             | 0.6             | 4     | 47.9           |
| 11  | 0.2             | 0.3             | 1     | 47.9           |
| 12  | 0.2             | 0.3             | 2     | 48.0           |
| 13  | 0.2             | 0.3             | 6     | 47.9           |
| 14  | 0.2             | 0.3             | 8     | 48.1           |
| 15  | 0.2             | 0.3             | 10    | 47.9           |
| 16  | 0.2             | 0.3             | 4     | <b>48.4</b>    |

Figure 1 plots the sensitivity results, varying one hyperparameter per panel.



**Fig. 1.** Sensitivity to one hyperparameter at a time, with the other hyperparameters fixed at their defaults.  $\lambda_{ref}$ ,  $\lambda_{bin}$ , and  $K_m$  denote the action-logit blending weight, completeness-residual weight, and number of retrieved supports, respectively.

## 6 Additional Ablations

Tables 5 and 6 give the fine-grained CAP and CPS loss-group ablations summarized in the main paper.

**Table 5.** CAP loss-group ablation on THUMOS14. AVG is over IoU thresholds  $\{0.1, 0.2, \dots, 0.7\}$ .

| Optimization Loss    |                         |                         |                          | mAP@IoU (%) |             |             |             |             |
|----------------------|-------------------------|-------------------------|--------------------------|-------------|-------------|-------------|-------------|-------------|
| $\mathcal{L}_{base}$ | $\mathcal{L}_{cap-mil}$ | $\mathcal{L}_{cap-loc}$ | $\mathcal{L}_{cap-disc}$ | 0.1         | 0.3         | 0.5         | 0.7         | AVG         |
| ✓                    |                         |                         |                          | 70.9        | 57.8        | 39.8        | 14.4        | 46.5        |
| ✓                    | ✓                       |                         |                          | 71.6        | 59.0        | 40.6        | 15.6        | 47.3        |
| ✓                    |                         | ✓                       |                          | 70.9        | 58.3        | 40.4        | 14.9        | 46.7        |
| ✓                    |                         |                         | ✓                        | 71.0        | 58.4        | 40.4        | 14.7        | 46.7        |
| ✓                    | ✓                       | ✓                       |                          | 71.6        | 59.1        | 40.6        | 15.7        | 47.3        |
| ✓                    | ✓                       |                         | ✓                        | 71.8        | 59.1        | 40.8        | 15.6        | 47.4        |
| ✓                    |                         | ✓                       | ✓                        | 71.1        | 58.4        | 40.4        | 14.7        | 46.8        |
| ✓                    | ✓                       | ✓                       | ✓                        | <b>71.9</b> | <b>59.4</b> | <b>40.9</b> | <b>15.8</b> | <b>47.6</b> |

**Table 6.** CPS loss-group ablation on THUMOS14. AVG is over IoU thresholds  $\{0.1, 0.2, \dots, 0.7\}$ .

| Optimization Loss    |                         |                         |                          | mAP@IoU (%) |             |             |             |             |
|----------------------|-------------------------|-------------------------|--------------------------|-------------|-------------|-------------|-------------|-------------|
| $\mathcal{L}_{base}$ | $\mathcal{L}_{cps-mil}$ | $\mathcal{L}_{cps-sup}$ | $\mathcal{L}_{cps-comp}$ | 0.1         | 0.3         | 0.5         | 0.7         | AVG         |
| ✓                    |                         |                         |                          | 70.9        | 57.8        | 39.8        | 14.4        | 46.5        |
| ✓                    | ✓                       |                         |                          | 71.6        | 59.0        | 39.9        | 14.2        | 46.9        |
| ✓                    |                         | ✓                       |                          | 71.9        | 59.4        | 39.7        | 14.6        | 47.0        |
| ✓                    |                         |                         | ✓                        | 71.8        | 59.3        | 40.0        | 14.5        | 47.1        |
| ✓                    |                         | ✓                       | ✓                        | 72.1        | <b>59.7</b> | <b>40.1</b> | 14.9        | 47.3        |
| ✓                    | ✓                       | ✓                       | ✓                        | <b>72.2</b> | 59.5        | 40.0        | <b>15.3</b> | <b>47.4</b> |

### 6.1 Module-Specific Reliability Signals

The main paper compares Base and CAPSNet on shared proposal-class pairs. Table 7 retains two checks that apply only to the proposed modules. The CAP gap is the mean action-dominance score of temporally correct pairs minus that of incorrect pairs. The CPS gap is the mean support confidence for present classes minus that for absent classes. Both gaps are positive and match the intended roles of CAP and CPS, but Base has no corresponding signals.

**Table 7.** Module-specific reliability signals on THUMOS14.

| Method  | $D_{TP} - D_{FP}$ | $\beta_{\text{present}} - \beta_{\text{absent}}$ |
|---------|-------------------|--------------------------------------------------|
| CAPSNet | +0.130            | +0.529                                           |